

Surakiat Kansa-ard

Junior AI Engineer

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ABOUT ME

Junior AI Engineer driven by a **Growth Mindset** and a passion for continuous learning, I specialize in developing **scalable, end-to-end AI solutions**. My expertise is built through in-depth projects, including **Multi-Stage Retrieval RAG** for data integrity, **Multi-modal RAG** supporting text, voice, and image search, and real-time **Computer Vision systems**. Additionally, I am proficient in managing the **Machine Learning lifecycle using MLOps** tools. With technical agility and multiple industry certifications, I am dedicated to building robust AI systems that deliver impactful results

PROJECTS

Multi-modal RAG Shop Assistant (LLMs + Multi-RAG) | <https://github.com/SurakiatP/Multi-modal-RAG>

End-to-end Multi-modal RAG system enabling product search through **text, voice, and image queries** with CLIP embeddings and optimized retrieval pipelines.

- Architected multi-modal RAG pipeline **achieving 0% hallucination, 93.5% answer relevance, and <1s response time** using CLIP (ViT-B/32) for image-text embeddings, Paraphrase-Multilingual-MPNet for text embeddings, and Qdrant vector database processing 12+ products.
- Implementing intelligent anti-hallucination protocols with strict source grounding, context completeness verification, and **LLM-as-a-Judge evaluation** using Opik platform with Groq API for optimal retrieval accuracy.
- Integrated Groq Cloud API with high-speed Qwen-3 (32B parameters) via LangChain for streaming responses, Faster-Whisper for voice input, and Thai language processing with edge-TTS multilingual synthesis.
- Deployed Gradio web application with dual interfaces (customer chatbot + admin panel) for comprehensive product management with async error handling.

Technologies: Python, Groq API, CLIP, Qdrant, Paraphrase-MPNet, LangChain, Gradio, FastAPI, Opik, Faster-Whisper, Computer Vision, NLP, TTS, STT, Thai Language Processing

Multi-Stage Retrieval RAG System (Hybrid Retrieval + Bilingual) | <https://github.com/SurakiatP/Multi-Stage-Retrieval-RAG>

End-to-end bilingual (Thai/English) Retrieval-Augmented Generation system for corporate policy documents featuring hybrid search, cross-encoder reranking, and temporal conflict resolution.

- Developed end-to-end RAG pipeline achieving 96.7% evaluation accuracy (**97.5% Faithfulness, 98.5% Answer Relevancy, 98.0% Policy Compliance**) using DeepEval framework with **LLM-as-a-Judge** (Groq Llama-3.3-70b) on 10 test cases covering policy updates and conflict resolution.
- Architected **hybrid retrieval system** combining BM25 sparse search (40%) with FAISS dense vector search (60%) using HuggingFace multilingual-e5-small embeddings, followed by BAAI/bge-reranker-base cross-encoder reranking (**Top-5**), **processing 1,200+ document chunks** with **sub-2s query latency**.
- Engineered bilingual document processing pipeline: Thai documents via EasyOCR rasterization + PyThaiNLP normalization/tokenization, English documents via Docling extraction, with **RecursiveCharacterTextSplitter (chunk_size=1200, overlap=300)** for semantic preservation.
- Implemented LLM-powered query expansion using scb10x/typhoon2.1-gemma3-4b generating Thai-English keyword pairs, and context-aware answer generation using qwen2.5:7b-instruct with **TOON tabular format** for structured context and automatic source citation.
- Deployed production FastAPI **REST API** with async endpoints, comprehensive logging, and Ollama integration for local LLM inference, reducing operational costs while maintaining <2s end-to-end response time.

Technologies: Python, LangChain, FAISS, Hugging Face Transformers, Ollama, PyTorch, FastAPI, PyThaiNLP, EasyOCR, DeepEval, Docker, RAG, NLP, Semantic Search, TOON Format

Uber Delivery Time Prediction MLOps (PoC to Demo MLOps Pipeline) | <https://github.com/SurakiatPluber-delivery-time-prediction-mlops>

End-to-end MLOps pipeline for real-time delivery time prediction with automated retraining, streaming inference, and production monitoring using Kafka, MLflow, Airflow, and Prometheus/Grafana.

- Architected production **MLOps pipeline achieving 60% faster training time** using **XGBoost (MAE: 3.2min, RMSE: 4.8min)** with automated **hyperparameter tuning** (RandomizedSearchCV, n_iter=10, cv=3), **experiment tracking** via MLflow, and **model versioning** with DVC for **reproducibility across 1,155 delivery records**.
- Engineered feature pipeline processing weather (OpenWeatherMap API), traffic, and temporal features with 25+ engineered variables including distance binning (4 categories), hour buckets (4 periods), severity mapping (weather: 0-3, traffic: 0-2), and one-hot encoding for categorical variables.
- Implemented **real-time streaming inference** using Kafka (2s intervals), consuming order events with FastAPI **serving predictions (<100ms latency)** and maintaining 95%+ prediction success rate with automatic model loading from MLflow registry (Production stage).
- Deployed **automated daily retraining pipeline** using Apache Airflow with DVC integration for data versioning, PostgreSQL backend for **metadata management**, and MLflow **model registration** ensuring continuous model improvement with zero downtime deployment.
- Established **production monitoring** stack with Prometheus metrics (prediction latency, success/failure rates, input distributions) and Grafana **dashboards tracking system** health, model performance, and resource utilization across containerized microservices (Docker Compose: 9 services).

Technologies: Python, Machine Learning, XGBoost, LightGBM, MLflow, Apache Airflow, DVC, Kafka, FastAPI, Docker, PostgreSQL, Prometheus, Grafana, Scikit-learn, Pandas, NumPy, MLOps, CI/CD

EDUCATION

King Mongkut's Institute of Technology Ladkrabang

Bachelor of Engineering | Major: Automation Engineer | Cumulative GPA: 3.30/4.00

PROFESSIONAL EXPERIENCE

IoT Solution Engineer Intern

Swift Dynamics (Bangkok, TH)

Contributed to R&D team developing Smart Camera Security solutions for manufacturing and industrial applications, implementing computer vision prototypes for real-time surveillance systems.

Accomplishments:

- Developed real-time anomaly detection system using OpenCV and computer vision algorithms on **Linux platforms (Ubuntu, Debian)**, reducing human monitoring errors by 75% through automated surveillance.
- Implemented 24/7 automated security monitoring pipeline processing 50+ camera feeds simultaneously, detecting suspicious activities with 88% accuracy and alerting security personnel in real-time.

SKILLS

Programming: Python, SQL, C/C++

ML/DL Frameworks: PyTorch, TensorFlow, scikit-learn, pandas, numpy

Generative AI & LLMs: LangChain, Hugging Face, OpenAI API, Ollama, RAG Systems, Prompt Engineering, NLP

Computer Vision: OpenCV, CLIP, OCR (EasyOCR, Tesseract), Image Classification

Vector Databases: FAISS, Qdrant, Chroma | **Embeddings:** BGE-m3, MPNet, multilingual-e5

Data Engineering: ETL Pipelines, Airflow, Kafka, Data Cleaning, PyThaiNLP

Backend & Deployment: FastAPI, Docker, REST API, CI/CD (GitHub Actions)

MLOps: MLflow, DVC, Model Versioning, Experiment Tracking

Visualization: Streamlit, Gradio, Matplotlib, Plotly, Grafana

Cloud: AWS (exposure) | **Version Control & Containerization:** Git, GitHub, Docker

Personal Traits : Curious, Collaborative, Eager to learn, Strong problem-solving, Willingness to explore new technologies